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Title: Manufacture of Pentaerythritol
Interim Report 1
Group 13

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1. Selected Chemical for Manufacture

Pentaerythritol

Pentaerythritol (or Tetramethylomethane) is an odourless, slightly sweet polyol organic compound with the formula $C_5H_{12}O_4$. It is a versatile building block for the preparation of many polyfunctionalized compounds.

99% MONO
PENTAERYTHRITOL

Figure 1 pentaerythritol[1]

The compound was first synthesized in the late 19th century and gained industrial importance due to its stability, high functionality and ability to form cross-linked polymers. Pentaerythritol is produced by aldol condensation of acetaldehyde and formaldehyde in the presence of a base catalyst, giving Penta erythrose, which then undergoes Cannizzaro reaction with formic acid [2].

2. Molecular Structure and Properties of Pentaerythritol

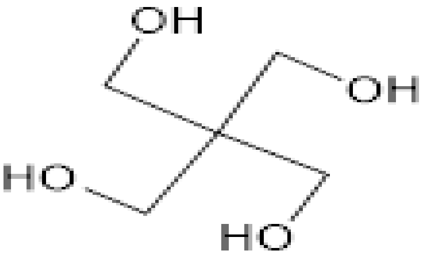

Figure 2 structure of pentaerythritol

- Pentaerythritol (C(H2OH)4) is a tetrafunctional polyol, meaning it has four hydroxyl (-OH) group attached to the central carbon atom. Its IUPAC name is 2,2-bis(hydroxymethyl)-1,3-propanediol.
- The core of the molecule consists of neopentane (tetrahedral) structure, where a central carbon atom is bonded to four hydroxymethyl (- CH2OH) group.

- Pentaerythritol has a central sp^3 hybridized carbon atom, resulting in a tetrahedral geometry with 109.5° bond angles [2].
- The four hydroxymethyl ($-CH_2OH$) groups are symmetrically arranged, leading to minimal steric hindrance. This reduces repulsion between the groups, contributing to the molecule's stability and uniform behaviour in physical and chemical interactions [2] .
- Each of the four –CH2OH group contains a reactive hydroxyl group, making pentaerythritol a polyhydric alcohol. These -OH group allow for esterification, etherification and polymerization reaction making PE useful in resins, plastics, and explosives.
- The presence of hydrogen bonding increases its melting point (260 C) and soluble in water.
- Electron-donating hydroxyl groups enable hydrogen bonding, thermal stability, and reactivity in ester formation.
- The tetrahedral carbon centre in pentaerythritol enables uniform molecular packing and strong crosslinking, enhancing mechanical strength, adhesion, and stability in coatings and adhesives.
- All bonds in pentaerythritol are sigma (σ) bonds, allowing free rotation around the C−C (1.548 Å) and C−O (1.43 Å) single bonds. This allows the molecule to adopt different conformations, giving it low-energy configurations.
- The carbon-oxygen bonds in pentaerythritol, especially those between the central carbon and the hydroxymethyl groups (−CH2OH), are relatively strong.
- It does not contain highly reactive groups, such as double bonds, aryl rings, or β hydrogen that might be more prone to thermal degradation.
  [3]

Table 1 properties of pentaerythritol

Properties	Description
IUPAC name	2,2-bis(hydroxymethyl)propane-1,3-diol
Molar mass	136.15 g/mol
Specific gravity	1.396
Hygroscopicity	Non- hygroscopic
Melting point	260 °C (533.15 K)
Boiling point	276 °C (549 K) at 30 mmHg
Dissociation constant (pka)	13.55
Specific Heat capacity	188.40 J/mol
Enthalpy of combustion	-2,800 kJ/mol
Enthalpy of formation	-920 KJ/mole
Enthalpy of fusion	17.64 KJ/mole
Enthalpy of sublimation	163KJ/mole
Enthalpy of vaporization	92 KJ/mole
Entropy	198 J/mole K

Gibbs free energy	-553 KJ/mole
Surface tension	32 ± 4 mN/m
Dynamic viscosity	Low (0.0000016 Pas, 391.81K)
pH	3.5-4.5
Solubility (Log10 of solubility in mol/l)	-0.44 in water 1.3 in DMSO, 1.176 in Butyl alcohol
Partition coefficient	-1.7 at 22℃ (hydrophilic)
Vapor pressure (@20°C)	0.00000008 mmHg (20°C)
Refractive index	1.548
Thermal stability	Decomposes at 200–270°C
Flash point	200°C, low flammability
Auto ignition point	390-490°C
Decomposition products	Carbon monoxide (CO), carbon dioxide (CO_2), formaldehyde, and acetic acid.
Polarity	Polar (due to hydroxyl groups)
Acid value	0.1 mg KOH/g
Iodine value	0
Environmental impacts	Low toxicity, bio-degradable
Electrical conductivity	Non- conductive
Thermal conductivity	Non- conductive

[2] [4]

3. Applications of Pentaerythritol:

The various applications of pentaerythritol are as follows:

- Alkyd resins: Alkyd resins are oil-modified thermoplastic organic polyesters that serve as film-forming agents in paints and clear coating. They are formed by polymerization of a polyol and an acid [5].It's used in synthetic varnishes, enamel, and industrial coatings. It provides durability in automotive, marine and architectural paints.
  Alkyd resins made from pentaerythritol are superior due to their,
    - higher hydroxyl groups compared to others, thereby in more cross-linking and increased performance. Tighter crosslinking creates a smoother surface, thereby increasing gloss [6]
    - Thermal stability, it prevents degradation at high temperature.

- Synthetic lubricants: Lubricants are substances that reduce friction on moving parts and reduce machine-driven wear [7]. Usage of pentaerythritol in lubricants can be more effective because,
    - Low viscosity so that it does not cause wear.
    - High thermal stability to withstand high temperatures without degrading.

  - High oxidation resistance so that it won’t form sludge or varnish when exposed to air and heat
  - Non-toxic and biodegradable in nature.
  - corrosion resistance and water resistance, it repels moisture and prevents emulsification to avoid performance loss
  - Quickly reacts with atmospheric oxygen and dries faster, with more Oxidation Induction Time (OIT) - how long the mixture can resist oxidation[8].

Figure 3 SO- soyabean oil RO- rosin esters[1][8]

- Rosin esters: Rosin esters are esters formed from natural resin obtained from pine trees. They are yellow and require the addition of modified additives to enhance stability. It is used as a tackifier in adhesives, and acts as a lubricant on the skin’s surface, giving the skin a smooth surface. Pentaerythritol can be a potential modifier due to its,

  - good thermal stability, making it less susceptible to deterioration due to environmental factors during industrial application or storage.
  - the four hydroxyl groups allow precise control over the esterification process, enabling the production of tailored rosin esters.
  - offers a good balance of hardness, flexibility, and stability [9]

- Antioxidants: They are substances that protect the cells from oxidation caused by free radicals. Pentaerythritol used in rubber industries inhibits oxidation of macromolecules and thereby prevents them from cracking and providing a prolonged service index. It’s all owed to neutralizing free radicals, enhancing the durability of polymers [10].

- Explosives: When pentaerythritol undergoes esterification with nitric acid, it produces compounds like nitro Penta (PETN) and Petrine. Pentaerythritol tetranitrate (PETN) is an explosive used as the primary ingredient in detonating fuses and as a component in some plastic explosives.

  In the human health field, both PETN and nitro-glycerine are used medicinally in the treatment of angina because of their shared vasodilatory action [11].
  It’s all due to its,
  - High Oxygen Balance, the molecular structure of pentaerythritol ensures efficient conversion to energy during combustion or detonation. ($C_5 H_8 N_4 O_{12}$)
  - The stability of pentaerythritol provides a safe base, while the nitrate groups enhance explosive power. High density and compact structure contribute to high energy release during detonation while maintaining stability [12].

- UV curing resins: UV-curing resins are made from Di pentaerythritol which is produced from pentaerythritol, with a wide range of applications. For example, they can be used to keep fine components in place when used by the electronics and semiconductor industries. In the printing industry, it can very effectively adhere to the products while drying quickly to increase printing efficiency. In addition to the industries, UV-curing resins are often used by the cosmetics and medical industries. White resins for teeth filling and the clear coating applied to nails at the end of a phototherapy nail session are both examples of UV curing resins produced from Di pentaerythritol. This is all due to the hydroxyl-rich structure facilitating rapid curing under UV light, enabling quick cross-linking in coatings [13].

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[1] The picture depicts that Pentaerythritol doped lubricants have high OIT

- Fire retardants: Pentaerythritol acts as a "charring agent" in fire retardant systems. The high carbon content facilitates the formation of a protective char layer on the material's surface when exposed to fire. The formed char acts as an insulating barrier, slowing down the combustion process and thereby improving the material’s fire resistance. When combined with other flame-retardant chemicals like ammonium polyphosphate (APP), it forms an intumescent flame-retardant system, which further expands upon heating to protect the material [14].

- Adhesives and sealants: Pentaerythritol is used in high –performance adhesives for aerospace and construction process.Pentaerythritol is preferred because of its strong cross linking, which improves bond strength and durability.

4. History of Manufacture of pentaerythritol (world and India):

Global Historical Development of Pentaerythritol

19th Century: Discovery & Early Research

1882 – Pentaerythritol was found as a by-product of the reaction between barium hydroxide and impure formaldehyde. It was recognized as a new four-valent alcohol with an unusual structure.
1891 – Tollens and Wigand successfully synthesized pentaerythritol by reacting formaldehyde and acetaldehyde with lime, resulting in regular crystal formation [4] .

20th Century: Industrial Advancements

1912 – German commercial production of PETN (a pentaerythritol derivative) started. It was extensively utilized as an explosive in World War I [2] .
1930 – Patent writings reported alterations to lime-based reactions, albeit the basic process remained the same.
1948 – Lime was replaced by caustic soda, with improved scalability and fewer impurities, thus increasing large-scale production's commercial feasibility.
1953 – A steady manufacturing process was established, a great industrial achievement in terms of efficiency and scalability.

21st Century: Modernization & Sustainability

2000 – Production factories became 99% automated, improving accuracy and minimizing human involvement.
2010 – Byproduct sodium formate was treated to yield reusable formic acid, enhancing sustainability in plants.
2020 – Yields were also optimized to 85–90%, with an increasing emphasis on eco-friendly production practices [15] .
2024 – The global market grew with investments in manufacturing infrastructure and technological updates, spurred by growing demand. The size of the global pentaerythritol market was USD 2.80 billion in 2023 and is projected to grow at a CAGR of 43.2% between 2024 and 2030 [16]$.

Indian Historical Development of Pentaerythritol

Pre-1950s: Imports Dependence

Pre-1950s – India was dependent on imports for pentaerythritol, mainly for the production of explosives and alkyd resins. [17]
1950–2000: Establishment & Modernization
1950 – Small-scale production of pentaerythritol commenced in India due to the need for alkyd resins in coatings and paints.
1960 – Kanoria Chemicals and Industries Ltd (KCIL) was incorporated in December 1960 and became the first indigenous company to manufacture pentaerythritol. The factory was at Ankleshwar, Gujarat [18] .
2000–2020: Technological Advancements & Market Expansion
2012–2017 – The Indian Chemical Industry Five-Year Plan encouraged domestic chemical plants, minimizing import dependency and augmenting indigenous pentaerythritol manufacturing units [19] .
2017 – Demand for pentaerythritol in India was 31,000 MTPA (metric tonnes per annum). Production capacity became 730 metric tonnes per month.
2021–Present: Growth & Sustainability

2021 – Asian Paints Ltd plans to double pentaerythritol manufacturing capacity at its Cuddalore factory, doubling production from 730 tons to 1,500 tons per month to fulfill increasing demand for paint and lubricant [20] .
2023 – Production volume of pentaerythritol in India was 15.6 thousand metric tons. India emerged as one of the world's largest chemical manufacturers, producing more than 80,000 products [21] .
2024 – Perstorp commissioned a new pentaerythritol plant at Sayakha, Bharuch, Gujarat. The plant has an annual capacity of 40,000 metric tonnes and uses renewable materials with hybrid electricity sources [22] **.**

5. Market Research on Manufacture Pentaerythritol:

PENTAERYTHRITOL MARKET – GLOBAL:

Pentaerythritol is majorly used to manufacture Alkyd resins, resins in manufacturing paints and coatings.

- Many countries like China, Hong Kong, and India have been actively boosting housing construction to provide homes for their citizens and launched several measures to push the construction of low-cost housing. Accordingly, the increasing residential building construction activities are poised to propel the demand for oil-based coatings, thus propelling the current studied market.

- The Asia-Pacific market is forecast to lead the pentaerythritol market throughout the forecast period. Increasing demand for pentaerythritol from plasticizers, adhesives and sealants, lubricants, coatings, and paints in the construction of residential housing and automobiles in nations such as China, Japan, and India is anticipated to propel the demand for pentaerythritol in this market.

- China is the largest car market in production and sales globally. As per OICA (The Organisation Internationale des Constructers d'Automobile), car production in China totalled 27.02 million units in 2022, and this growing production of automotive vehicles is expected to propel the lubricant market, hence propelling the demand for pentaerythritol.

- Likewise, in India, growth in demand is being witnessed from the industrial and automotive sectors for lubricants. For example, in March 2023, ExxonMobil said it would invest close to USD 110 million to construct a manufacturing facility for lubricants at the Maharashtra Industrial Development Corporation's Isambe Industrial Area in Raigad. Therefore, growth in the volume of production of lubricants will spearhead the market.
- Owing to the above-mentioned factors, the market for pentaerythritol in the Asia-Pacific region is projected to grow significantly during the forecast [15].

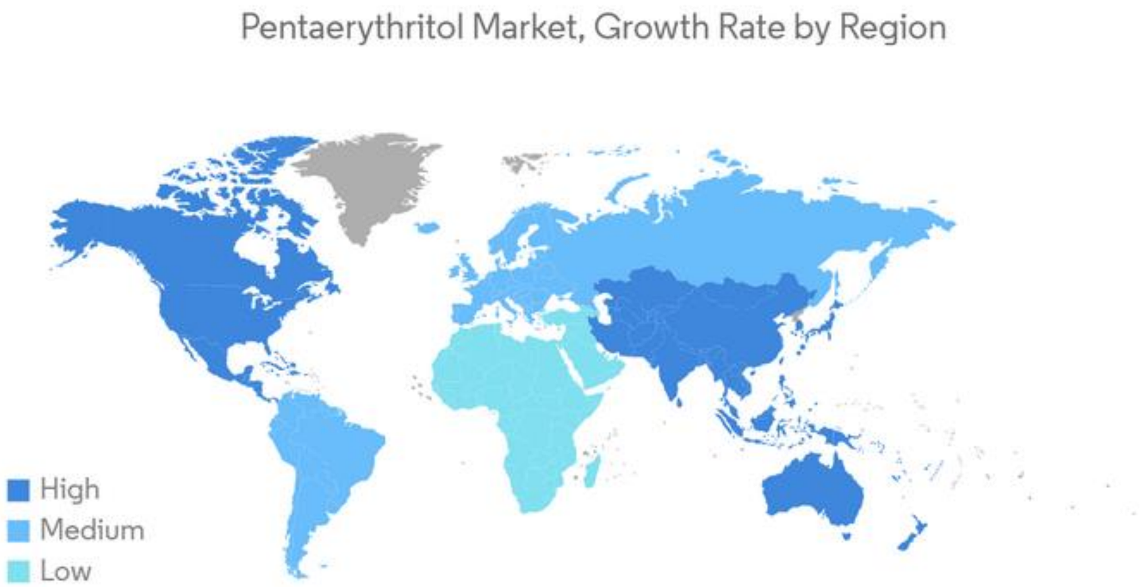


Figure 4 pentaerythritol market and growth rate by region[15]

Study Period	2022-2027
Market Volume (2025)	733.90 kilotons
Market volume (2027)	816.85 kilotons
CAGR (2024-30)	5.50%
Fastest Growing Market	Asia Pacific
Largest Market	Asia pacific
Market Concentration	High

Table 2 world market of pentaerythritol[16]

The global pentaerythritol market generated a revenue of USD 2,800.0 million in 2023 and is expected to reach USD 34,612.0 million by 2030. In terms of revenue, North America was the largest revenue-generating market in 2023. Asia Pacific region accounted for 24.5% of the global pentaerythritol market in 2023 and is the fastest growing regional market and is projected to reach USD 8,680.7 million by 2030. The Pentaerythritol Market size is estimated at 733.90 kilotons in 2025, and is expected to reach 816.85 kilotons in 2027, growing at a CAGR of greater than 5.5% during the forecast period (2022-2027) [17].

Pentaerythritol synthesis costs depend on formaldehyde and acetaldehyde prices. These input prices can fluctuate, changing market dynamics and altering manufacturers' and suppliers' pricing and profit margins. Pentaerythritol market concentration is more inclined towards consolidated as market is being dominated by some 5-10 major players. Some of the major players of Pentaerythritol in market include:

- Ercros SA

- KH Chemicals
- Perstorp
- Chemanol
- HUBEI YIHUA CHEMICAL INDUSTRY CO., LTD [15]

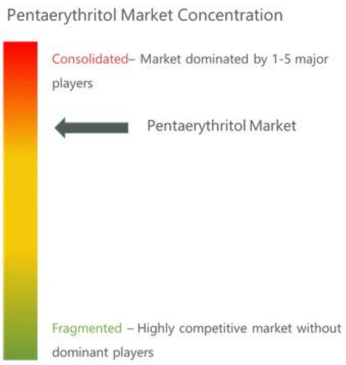

Pentaerythritol Market Concentration
Consolidated– Market dominated by 1-5 major players
Pentaerythritol Market
Fragmented – Highly competitive market without dominant players

Figure 5 pentaerythritol market concentration[17]

PENTAERYTHRITOL MARKET - INDIA:

The production volume of pentaerythritol across India in fiscal year 2023 amounted to 15.6 thousand metric tons. Turning out over 80 thousand products, the south Asian country is one of the largest producers of chemicals in the world.

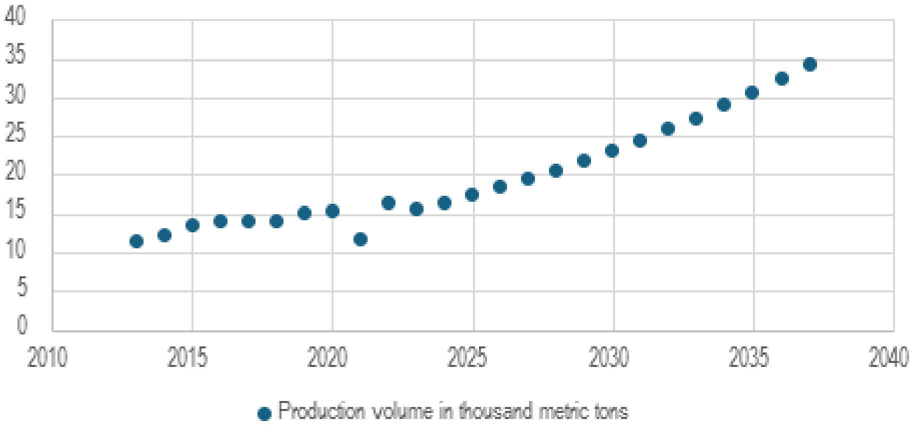


Figure 6 production volume in thousand metric tons vs. year of production [18]

The India Pentaerythritol market was valued at $89.5 Million in 2023, and is projected to reach $157.28 Million by 2033 growing at a CAGR of 5.80% from 2024 to 2030. Alkyd Paints segment is expected to be the highest contributor to this market, with $38.8 Million in 2023, and is anticipated to reach $67.9 Million by 2033, registering a CAGR of 5.79%. Alkyd Varnishes segment is anticipated to reach $18.2 Million by 2033 with the highest CAGR of 9.00%. Alkyd Paints and Alkyd Inks segments collectively expected to account for about 56.6% share of the India Pentaerythritol market in 2023, with the former constituting around 43.4% share. Alkyd Varnishes and Radiation Cure Coatings segments are expected to witness significant growth rates at a CAGR of 9.00% and 7.77% respectively, during the forecast period. Presently, the share of these two segments is estimated to be around 14.8% in the overall India Pentaerythritol market in 2023 and is anticipated to reach 18.2% by 2033 [19].

IMPORTS AND EXPORTS:
In 2022 Pentaerythritol was the world's 3012th most traded product (out of 4,648)

The top importers of **Pentaerythritol** were United States ($95.2M), India ($33.8M), South Korea ($24.5M), Japan ($22.2M), and Brazil ($19.4M) [20].

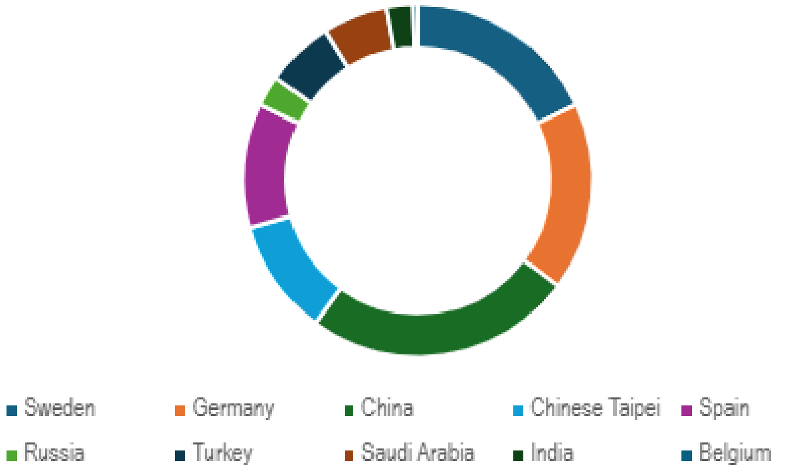


Figure 7 exports of pentaerythritol[20]

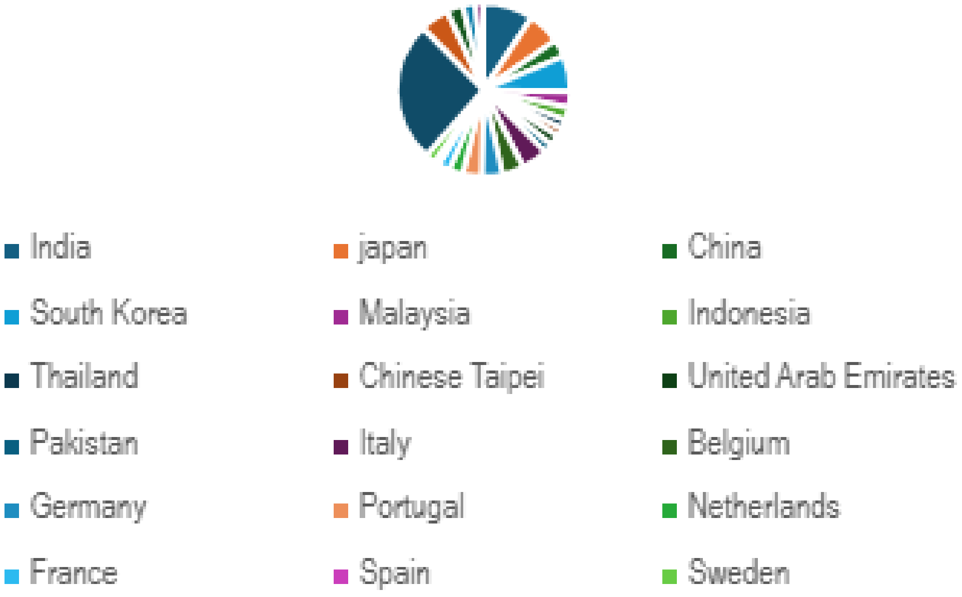


Figure 8 imports of pentaerythritol[20]

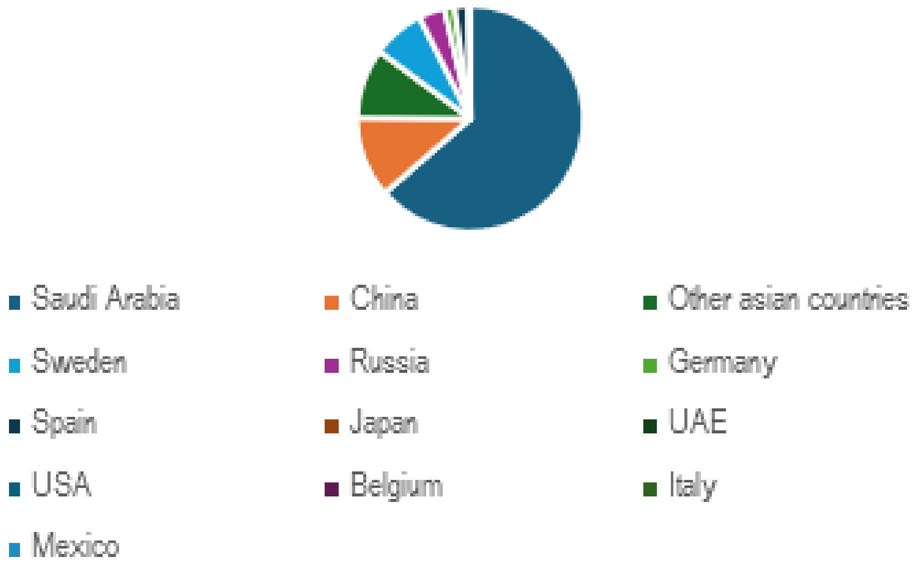


Figure 9 imports of pentaerythritol from other countries to India in kg[20]

India imports 32,042.73(1000 USD) in 2023, in quantity 23,174,000 kg of pentaerythritol from other countries. We import a major part from Saudi Arabia (USD 13,540,970) and China (5,629,110 USD) as mentioned above in the diagram.

India's Anti-Dumping Measures**:** In May 2023, India imposed anti-dumping duties on pentaerythritol imports from China, Saudi Arabia, and Taiwan to protect domestic manufacturers from unfair pricing practices. The duty on imports from China was set at USD 345.15 per tonne. This measure is valid for five years and aims to ensure fair competition for Indian producers.

In the case of Pentaerythritol imports from China, the revenue department has imposed anti-dumping duty of $ 345.15 per tonne. For Pentaerythritol imports from Saudi Arabia, the anti-dumping duty has been pegged at $ 300.15 per tonne. In the case of Pentaerythritol imports from Taiwan, the revenue department has imposed anti-dumping duty of $ 499.01 per tonne. Pentaerythritol is classified under the Harmonized System (HS) code 2905 42 90 in India's tariff schedule [21].

Country of Origin	**Country of Export**	**Amount (MT)**	**Currency**
China PR	Any country including China PR	345.15	USD
Any other country than China	China PR	345.15	USD
Saudi Arabia	Any country including Saudi Arabia	300.15	USD
Any other country than Saudi Arabia.	Saudi Arabia	300.15	USD

Taiwan	Any country including Taiwan	499.01	USD
Any country other than Taiwan	Taiwan	499.01	USD

Table 3 Anti-dumping,Tarrif item.no -29054290 [22]

Environmental Regulations:

This regulation by Environmental Protection Agency (EPA) establishes exemptions from the requirement of a tolerance for residues of pentaerythritol tetrakis (3-(3,5-di-tert-butyl-4-hydroxyphenyl) propionate) (CAS Reg. No. 6683-19-8) under 40 CFR 180.910 and 180.930 when used as an inert ingredient (antioxidant/stabilizer) in pesticide formulations applied to growing crops and raw agricultural commodities after harvest at a maximum concentration of 5% by weight in the formulation and applied to animals at a maximum concentration of 3% by weight in the formulation, respectively. BASF Corporation submitted a petition to EPA under the Federal Food, Drug, and Cosmetic Act (FFDCA), requesting establishment of these exemptions from the requirement of a tolerance. These regulations eliminate the need to establish a maximum permissible level for residues of pentaerythritol tetrakis (3-(3,5-di-tert-butyl-4-hydroxyphenyl) propionate) for these uses.
The Environmental Protection Agency (EPA) oversees the regulation of chemicals under the Toxic Substances Control Act (TSCA). Specific chemicals, including certain derivatives of pentaerythritol, may be subject to exemptions or specific regulatory requirements [23]

Company Name	**Location**	**Production Capacity/plant**	**Total production capacity**
Jiangsu Ruiyang Chemicals[24]	Jiangsu	20,000	50,000
	Chifeng	20,000	
	Guiyang	10,000	
Henan Pengcheng Group[25]	Puyang Pengxin	35,000	60,000
	Puyang Pengcheng	15,000	
	Puyang Kenko Pengcheng	10,000	
Hubei Yihua Chemicals[33]	Hubei,China	110000	110000
Boarding Guoxiu Chemicals[34]	Hubei,China	20,000	20,000

Table 4 company and its production in the global market

Company Name	**Location**	**Production(tons/yr)**

Kanoria chemicals [28]	Ankleshwar, Gujarat	30000
	Visakhapatnam, Andhra Pradesh	
Asian paints[35]	Cuddalore,Tamil Nadu	8760
Vizag[36]	Vishakhapatnam, Andhra Pradesh	15000
Perstop[37]	Sayakha,India	40000

Table 5 company and its production in India

Item	Grade 98%	Grade 95%	Grade 93%
Mono-pentaerythritol Content, %, ≥	98	95	93
Hydroxyl Group, % ≥	49	48	47
Heating Loss, % ≤	0.2	0.5	0.5
Ash Content, % ≤	0.05	0.1	0.1
Final Melting Point, °C ≥	250	230	230
Color, (Pt Co) ≤	100	200	200
Appearance	White crystalline powder without any visible foreign matters.		

Table 6 grades of pentaerythritol available in market[38]

The pentaerythritol market can be segmented based on product type, application, and region:

- By Product Type: Pentaerythritol is available in various grades, such as technical grade and nitrated grade. Technical grade is commonly used in paints, coatings, and adhesives, while nitrated grade is suitable for applications like explosives and fire-retardant materials.
- By Application: Major applications include alkyd resins, lubricants, plasticizers, flame retardants, and adhesives. Alkyd resins hold a significant market share due to their widespread use in paints and coatings. Lubricants and flame retardants are also growing segments driven by environmental and safety requirements.
- By Region: The market is divided into Asia-Pacific, North America, Europe, Latin America, and the Middle East & Africa, with each region showing distinct demand patterns based on local industry and regulatory factors.

These segments demonstrate the broad utility of pentaerythritol in multiple applications, each catering to specific industry needs [39].

6. Basis for Manufacturing Process with Justification

The global production capacity of pentaerythritol was estimated to be around 6,25,000 tons/yr in 2022.

CAGR = 5.5%
The current demand would be,
n = 2025 – 2022 = 3
Forecast Demand = (Current Demand) * (1+ (CAGR/100)) ^n
Forecast Demand (2025) = (6,25,00 tons/yr) * (1 + 0.055) ^3 = 733900 tons/year (approx.)
The forecast demand in the year 2027 would be,
n = 2027-2025 = 2
Forecast Demand = (733900 tons/yr) * (1 + 0.055) ^2 = 816849 tons/yr (approx.)
Gap in Production = Forecast Demand – Current Demand
Gap in Production = 816849 - 733900 = 82,949 tons/year ~ 83,000 tons/yr
But, according to production data in India,
Perstop (highest) = 40,000 tons/yr
Asian paints (lowest) = 8760 tons/yr
Since the gap is more than the current highest production, we take the basis to be the current highest production rate in India.
Basis = 40,000 tons/yr [24]

7. Selection of Synthesis Route and Justification for Pentaerythritol

Various methods employed for production of Pentaerythritol:

1) Using Sodium hydroxide (NaOH)

$$4HCHO + CH_3CHO + NaOH \rightarrow C(CH_2OH)_4 + NaHCOO$$

Operating temperature – 15 - 48°C

Yield (%) – 85-90

Advantages:

1. High yield pentaerythritol
2. Easy removal of sodium ions, Byproduct – Sodium formate can be used to produce formic acid
3. Catalyst supply is accessible
4. Ease of separation
5. Sodium formate has low toxicity to the biosphere

Disadvantages:

1. Sodium hydroxide is highly corrosive to metals and needs extra supervision on the equipment..

2) Using Calcium Hydroxide (Ca(OH)2)

$$4HCHO + CH_3CHO + \frac{1}{2}Ca(OH)_2 \rightarrow\ C(CH_2OH)_4 + \frac{1}{2}Ca(HCOO)_2$$

Operating temperature – 20- 60 °C

Yield (%) – 86-90

Advantages:

- Calcium hydroxide is cheaper than sodium hydroxide, Less usage of catalyst
- Calcium formate has low toxicity to the biosphere
- The byproduct, calcium formate, is insoluble in water and precipitates during the reaction. This requires an additional filtration step to separate the solid calcium formate, adding to the complexity and time needed for separation.

Disadvantages:

1. The high acidity can cause the equipment to corrode
2. Exposure to calcium hydroxide can cause severe skin irritation, chemical burns, blindness, lung damage, and rashes.

3) Using calcium hydroxide, sulphuric acid and methanol

$$4HCHO + CH_3CHO + \frac{1}{2}Ca(OH)_2 + \frac{1}{2}H_2SO_4 + CH_3OH \rightarrow\ C(CH_2OH)_4 + \frac{1}{2}CaSO_4 + HCOOCH_3 +\ H_2O$$

Operating temperature – 18- 41 °C

Yield (%) – 80

Advantages:

1. Minimising side reactions
2. High yield of pentaerythritol
3. High purity of pentaerythritol

Disadvantages:

1. Requires additional processes/ equipment, Forming azeotropic composition through the reaction of ester and water which will require advanced distillation column
2. additional raw materials
3. Difficult in the removal of Ca ions
4. Methyl formate formed during the reaction reacts with water forms a corrosive acid
5. The calcium sulphate formed can cause toxicity to fish
6.

4) Using calcium hydroxide, sodium carbonate, methanol

$$4HCHO + CH_3CHO + \frac{1}{2}Ca(OH)_2 + \frac{1}{2}CaCO_3 \rightarrow C(CH_2OH)_4 + \frac{1}{2}CaCO_3 + NaHCOO$$

Operating temperature – 20- 60 °C

Yield (%) - <85

Advantages:

1. Calcium hydroxide is cheaper than sodium hydroxide, Less usage of catalyst
2. Low potential to affect the biosphere

Disadvantages:

1. The process requires a large number of equipment including two pan filters
2. Two separate processes are required for the removal of calcium carbonate and sodium formate
3. High possibilities of the occurrence of side reaction

The sodium hydroxide (NaOH) pathway is the best method for pentaerythritol production due to its high yield (85- 90%), easy removal of sodium ions, and useful sodium formate byproduct. While NaOH is corrosive, its advantages like high yield, catalyst availability, and environmental safety outweigh its drawbacks. The other pathways require additional purification steps and pose toxicity risks.

Therefore, the NaOH based process is been selected for its efficiency, simplicity and sustainability.[40]

8. Process Flow Chart for the Selected Synthesis Route

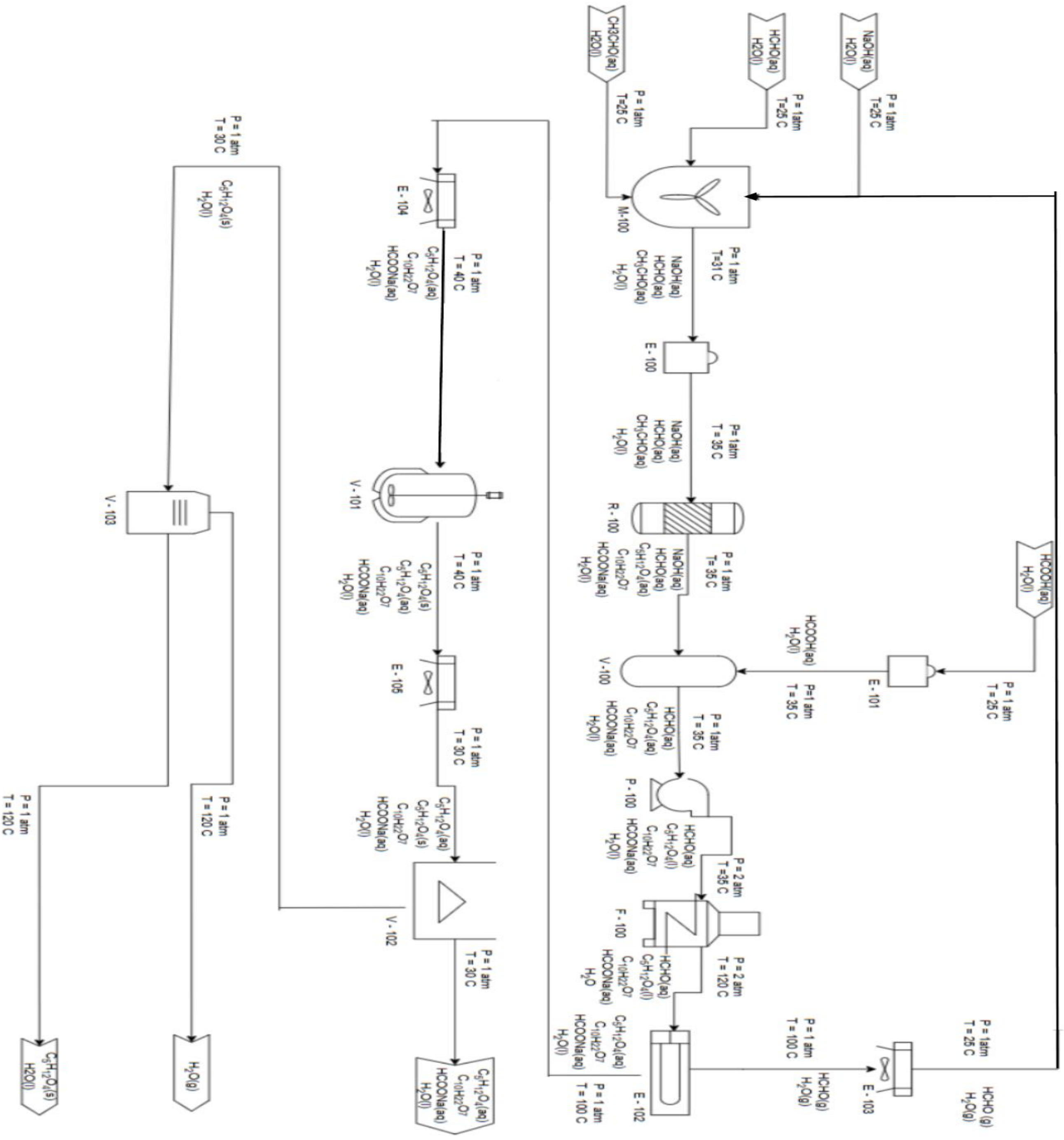

NaOH(aq)
H2O(l)
P = 1atm
T=25 C
HCHO(aq)
H2O(l)
P = 1atm
T=25 C
CH3CHO(aq)
H2O(l)
P = 1atm
T=25 C
M-100
E - 100
R - 100
V - 100
P - 100
F - 100
E - 102
E - 103
E - 101
HCOOH(aq)
H2O(l)
E - 104
V - 101
E - 105
V - 102
V - 103
HCHO(g)
H2O(g)
P = 1 atm
T = 120 C

Figure 10 pfd for pentaerythritol manufacturing [40]

Process description:

Raw Material Preparation and Feeding

The production process starts with feeding of the key raw materials into the Mixer:

- Formaldehyde aqueous solution (25 %)
- Acetaldehyde (99%)
- Sodium Hydroxide (16%) - A base catalyst, for initializing aldol condensation.

The mole ratio of formaldehyde to acetaldehyde generally used is 5:1. A ratio of 1:2 mole hydroxide ion per mole of acetaldehyde appears to be the optimum amount of condensing agent. These materials are fed into agitated Mixing Tank (M-100) to ensure uniform distribution, at 1 atm and 25 C, the temperature is low enough to prevent any appreciable loss of acetaldehyde by evaporation (20.2 C). An immediate exothermic reaction occurs which increases the temperature from 25 C to 31 C. This reaction occurs due to spontaneous initiation of aldol condensation in presence of strong base catalyst.

Boiler:

The mixture is preheated using heat exchanger (E-100) to reach 40 C before entering the reactor, for the following reasons:

- Aldol condensation and Cannizaro reaction require a controlled temperature for optimal conversion.
- If the reaction mixture is not heated uniformly, undesirable polymeric byproducts may form, the boiler ensures that the reactants are uniformly preheated before entering the main reactor (R – 100).

Reactor(R-100):

Feed Acetaldehyde, Formaldehyde and sodium hydroxide solution into a stirred reactor. To achieve optimal conversion, a cooling Jacket or external cooling system is employed to maintain temperature at 40 C to avoid the side reactions, typically chilled water or brine is used as cooling medium.

A series of Aldol condensation (Base – Catalysed) reaction occur forming a trimethoxy acetaldehyde as intermediate. Here the intermediate reacts with another molecule of formaldehyde, this step is repeated more than twice progressively building the Carbon chain.

Simultaneously, a Cannizaro reaction takes place where the intermediate compound (Pentaerythrose) it reacts with fourth molecule of formaldehyde, resulting in the formation of pentaerythritol and sodium formate.

Pentaerythritol is made from acetaldehyde and formaldehyde in presence of an alkaline condensing agent. Initial reaction is three successful molecules of formaldehyde added to one molecule of acetaldehyde by aldol condensation

$$CH_3CHO + HCHO \rightarrow HOCH_2 - CH_2CHO$$

(Acetaldehyde reacts with formaldehyde to form 3-hydroxypropanol)

$$HOCH_2 - CH_2CHO + HCHO \rightarrow (HOCH_2)_2CHCHO$$

(formation 2,2-bis (hydro methyl) propanol)

$$(HOCH_2)_2CHCHO + HCHO \rightarrow (HOCH_2)_3CCHO$$

Formation of Trihydroxymethylpropanol (THMP)

These reactions are truly catalytic and consume no base. They are also reversible the equilibria of all the above reactions are displaced continuously to the right by the second possible reaction between two aldehydes in the presence of a base (NaOH) that is crossed Cannizaro reaction and which is not reversible

Pentaerythritol is formed as the main product, and Sodium formate (NaHCOO) is formed byproduct.

Most of the side reactions occur due to formaldehyde and acetaldehyde reacting with each other. Reactions of acetaldehyde with itself are rapid. However, these reactions can be minimized by maintaining formaldehyde concentration throughout

Formaldehyde can react with itself to form sugar like products which eventually caramelize to import colour and Odor to the products. The self-Cannizaro of formaldehyde to for CO2 can be minimized by avoiding high concentration of formaldehyde and high temperature. The desired Cannizaro does not progress rapidly below 40 degree and pH 10.

Neutralizing Tank (V – 100):

To neutralize the excess alkali (NaOH) and to reduce the impurity formed we add Formic acid, to prevent any side reaction to occur. The mixer is adjusted to slightly acidic pH around 6-7. But the temperature and pressure are still maintained at 35 C and 1 atm.

Before adding the formic acid to the Neutralizing Tank (V-100) we heat it to 35 C to maintain the temperature at V-100 using E-100.

Maintaining Slightly Acid pH:

- Ensures the precipitation of unwanted salts
- Prevents the formation of excess Sodium based impurities.
- Reaction in Neutralization Tank.

$$NaOH + HCOOH \rightarrow H_2O + NaHCOO$$

Pump (P-100):

Pump is used to increase the pressure of the mixture to 2 atm, increasing pressure increases boiling point of water and other components, also prevents the vaporization of volatile compounds like acetaldehyde, High pressure aids in efficient separation of reactants. Improves process efficiency and energy utilization, ensuring uniform distribution of heat.

Fired Heater (F-100):

It is used to increase the temperature of the components to 120 C maintaining the same pressure (2 atm). Gaseous formaldehyde is recovered for recycling.

Evaporation Unit (E-102):

The temperature inlet at 120 C is set to allow water /formaldehyde to get vapourised.

Higher pressure at 2 atm is to set the outlet pressure of evaporator to 1 atm, the higher pressure to lower pressure aids in separation of pentaerythritol from reaction byproducts. At 100 C and

1 atm ,the ratio of solute to solvent is at 0.39 :1 an optimal feed ratio as a liquor before entering the crystallizer. The Formaldehyde vapor is condensed and recovered back to use at E- 103 where the temperature of formaldehyde is reduced to 25 C.

Heat exchanger (E-104):

In the previous step we evaporate the aqueous solution to concentrate the pentaerythritol but here, it's used to reduce the temperature of the components to 40 C to aid in the crystallization process.

Crystallizer (V-101):

Crystallizer is operated at 40 C and 0.6 atm.The temperature of crystallizer must not be too low or too high may result in least crystal formed, too low may cause the sodium formate to form. The temperature of formation of sodium formate is at ambient T in accord to the formic acid and water content. Thus, the T of crystallizer must be above 25 C. P under vacuum is used to produce pentaerythritol slurry.

We use agitated vessel to ensure even temperature distribution and prevent local supersaturation, prevents excessive agglomeration and formation of non – uniform crystals.

Centrifugal filter (V-102):

The outlet from the crystallizer is brought to ambient temperature with the help of heat exchanger (E –105), we filter out the solid pentaerythritol crystals from mother liquor using Centrifugal filter. The centrifugal filters are performed at high speed, to make the filtration process efficient.

Dryer (V – 103):

Dryer separates solid pentaerythritol from the mother liquor, Moisture control affects the stability and handling of the final product. The temperature is set to 120 C which vaporizes most of the water within the crystals. The temperature increase is brought by direct contact of the slurry with heated ambient air or superheated steams is used. Direct contact with heated gas ensures rapid evaporation of water, efficient removal of moisture, and we can adjust the system to prevent overheating.

By product formed 0.15 of di pentaerythritol is produced for 1 mol of pentaerythritol is produced [40].

DETAILED EXPLANATION OF THE PROCESS:

Table *7* Heat exchanger details and its temperature change[40]

Location of heat exchanger	**Purpose**	**Inlet temperature**	**Outlet temperature**	**Heat exchange medium**
Heat exchanger(E-100) Before reactor(R-100)	Preheats the reaction mixture before entering the reactor to ensure uniform temperature and prevent polymeric byproducts	31°C	35°C	Hot water~45°C
Heat Exchanger (E-101) Before neutralizing tank(V-100)	Ensure the formic acid is at the same temperature as the reaction mixture to prevent sudden colling effects in neutralizing tank	~25°C	~35°C	Steam condensate at ~40°C
Fired heat exchanger(F-100) Before evaporation unit(E-102)	Increases temperature for evaporation and separation of formaldehyde gas.	35°C	120°C	Direct fired heater (gas or fuel combustion)
Heat exchanger (E-103) After evaporation unit(E-102)	Condense and recovers the formaldehyde gas for reuse	120°C	25°C	Chilled water (at ~10°C)
Heat exchanger(E-104) Before crystallizer(V-101)	Lowers temperature to facilitate crystallization of pentaerythritol	~120°C	~40°C	Cooling water at 25°C
Heat exchanger(E-105) Before centrifugal filter(V-102)	Reduces the temperature to prevent unwanted precipitation before	~40°C	~25C	Chilled brine(~25C)

	centrifugal filitration			

9. Bill of Materials based on raw materials

Table *8* bill of materials for pentaerythritol production

S.no	Chemicals required	Rate/kg	Kg required/hr	Total amount (for basis of 133333.33kmol/hr of pentaerythritol)
1.	formaldehyde	₹ 15 (purity 37%) \kg	163218.6681	₹ 2448280.022
2.	acetaldehyde	₹ 850\kg	47931.88221	₹ 40742099.88
3.	Sodium hydroxide	₹ 65\kg	60103.64235	₹ 3906736.753
4	Formic Acid	₹ 93\kg	17354.92673	₹ 1614008.186

10. Material Balances for the Entire Process to produce Pentaerythritol

From Basis = 133333.33 kg of Pentaerythritol
Which is 979.3120088 kmol of Pentaerythritol
Water lost due to crystallization is taken as 6% as mentioned in the literature.

Yield	0.9	Actual moles of pentaerythritol/moles of acetaldehyde fed	
		Moles of acetaldehyde fed (in kmol/hr)	1088.124454
Selectivity	0.86	Moles of pentaerythritol produced/Total product produced	
		Total product produced (in kmol/hr)	1138.734894
Conversion	0.90	Moles of acetaldehyde consumed/Moles of acetaldehyde fed	
		Moles of acetaldehyde consumed (in kmol/hr)	979.3120088

OVERALL MATERIAL BALANCE:

Input	Feed Ratio	Mass Percentage	Molecular weight	Molar flow rate(Kmol/hr)	Mass flow rate(kg/hr)	Pure Mass flow rate(kg/hr)	Water Mass Flow rate (kg/hr)	
Formaldehyde	5	25%	30	5495.028494	288488.9959	164850.8548	123638.1411	
Acetaldehyde	1	99%	44.05	1099.005699	48900.20306	48411.20103	489.0020306	Total Water mass flow rate produced (kg/hr)
Sodium Hydroxide	1.2	16%	46.03	1318.806839	379404.2424	60704.67878	318699.5636	538234.3729
Water			18.01	29885.30666	0	538234.3729		
Formic Acid	1	34.65%	46.03	1099.005699	145994.8985	50587.23231	95407.66614	
				Total mass input		862788.3398	kg\hr	
Output	Ratio	Molecular weight	Molar flow rate (kmol/hr)	Mass flow rate(kg\hr)				

Total mass input inkg/hr 862788.3398 862788.3398

				Total mass input
Output	Ratio	Molecular weight	Molar flow rate (kmol/hr)	Mass flow rate(kg\hr)
Pentaerythritol	1.00	136.15	979.3120088	154862.7049
Formaldehyde	1.00	30	1577.780459	47333.41376
Water		18.01	28689.89439	516704.998
Sodium formate+Unreacted catalyst		68	1318.806839	89678.86502
Di pentaeryhtritol	0.15	254.31	163.2186681	41508.13949
Acetaldehyde		44.05	108.8124454	4793.188221
			Total mass flow rate	854881.3094
			Total Product produced	1142.530677

			Error in mass percentage	0.916450777

Total mass output = 854881.3094 kg/hr

Error= $\frac{862788.3398 - 854881.3094}{862788.3398} * 100$
= 0.91645077%

INDIVIDUAL MATERIAL BALANCE:
Mixer (MX – 100)

MIXER (M - 100)				
	INPUT		OUTPUT	
Components	Mass flow rate (kg/hr)	Molar flow rate(kmol/hr)	Mass flow rate(kg/hr)	Molar flow rate(Kmol/hr)
NaOH	60704.67878	1318.806839	60704.67878	1318.806839
Formaldehyde	164850.8548	5495.028494	164850.8548	5495.028494
Acetaldehyde	48900.20306	1110.106766	48900.20306	1110.106766
Water	538234.3729	29885.30666	538234.3729	29885.30666
Total flow rate	812690.1095	37809.24875	812690.1095	37809.24875
Input-output	0	(No accumulation)		

HEAT EXCHANGER /BOILER(E – 100)

	INPUT		OUTPUT	
Components	Mass flow rate (kg/hr)	Molar flow rate(kmol/hr)	Mass flow rate(kg/hr)	Molar flow rate(Kmol/hr)
NaOH	60704.67878	1318.806839	60704.67878	1318.806839
Formaldehyde	164850.8548	5495.028494	164850.8548	5495.028494
Acetaldehyde	48900.20306	1110.106766	48900.20306	1110.106766
Water	538234.3729	29885.30666	538234.3729	29885.30666
Total flow rate	812690.1095	37809.24875	812690.1095	37809.24875
Input-output	0	(No accumulation)		

REACTOR (R 100)

	INPUT	OUTPUT

Components	Mass flow rate (kg/hr)	Molar flow rate(kmol/hr)	Mass flow rate(kg/hr)	Molar flow rate(Kmol/hr)
NaOH	60704.67878	1379.65179		
Formaldehyde(HCHO)	164850.8548	5495.028494	32970.17096	1099.005699
Acetaldehyde(CH_3CHO)	48900.20306	1110.106766		
Water	538234.3729	29885.30666	538234.3729	29885.30666
Pentaerythritol ($C_5H_{12}O_4$)	0	0	154862.7049	1137.441828
Sodium formate(HCOONa)+Unreacted Sodium Hydroxide	0	0	60704.67878	892.7158644
Unreacted Acetaldehyde			2688.776264	61.03918874
Di Pentaerythritol	0	0	23229.40574	91.35364849
Total flow rate	812690.1095	37870.09371	812690.1095	33166.86288
Input-output	4.39235E-07			

NEUTRALISING TANK (V 100)

	INPUT		OUTPUT	
Components	Mass flow rate (kg/hr)	Molar flow rate(kmol/hr)	Mass flow rate(kg/hr)	Molar flow rate(Kmol/hr)
Sodium formate(HCOONa)+Unreacted Sodium Hydroxide	60704.67878	892.7158644	206699.5772	1991.721563
Pentaerythritol (C5H12O4)	154862.7049	1137.441828	154862.7049	1137.441828
Water	538234.3729	29885.30666	538234.3729	29885.30666
Formaldehyde(HCHO)	32970.17096	1099.005699	32970.17096	1099.005699
Formic Acid(HCOOH)	145994.8985	1099.005699	0	0
Di Pentaerythritol	23229.40574	91.35364849	23229.40574	91.35364849
Total flow rate	955996.2317	34204.82939	955996.2317	34204.82939

PUMP (P 100)

	INPUT		OUTPUT	
Components	Mass flow rate (kg/hr)	Molar flow rate(kmol/hr)	Mass flow rate(kg/hr)	Molar flow rate(Kmol/hr)
Formaldehyde(HCHO)	32970.17096	1099.005699	32970.17096	1099.005699
Pentaerythritol (C5H12O4)	154862.7049	1137.441828	154862.7049	1137.441828
Sodium formate(HCOONa)	206699.5772	54	206699.5772	54
Water	154862.7049	8598.706547	154862.7049	8598.706547
Di Pentaerythritol	23229.40574	91.35364849	23229.40574	91.35364849
Total flow rate	572624.5638	10980.50772	572624.5638	10980.50772

Input-output	0			

FIRED HEATER (F 100)

	INPUT		OUTPUT	
Components	Mass flow rate (kg/hr)	Molar flow rate(kmol/hr)	Mass flow rate(kg/hr)	Molar flow rate(Kmol/hr)
Formaldehyde(HCHO)	32970.17096	1099.005699	32970.17096	1099.005699
Pentaerythritol (C5H12O4)	154862.7049	1137.441828	154862.7049	1137.441828
Sodium formate(HCOONa)	206699.5772	3039.252716	206699.5772	3039.252716
Water	154862.7049	8603.483606	154862.7049	8603.483606
Di Pentaerythritol	23229.40574	91.35364849	23229.40574	91.35364849
Total flow rate	572624.5638	13970.5375	572624.5638	13970.5375
Input-output	0	(No Accumulation)		

EVAPORATOR (E 102)

	INPUT		OUTPUT		
Components	Mass flow rate (kg/hr)	Molar flow rate(kmol/hr)	Mass flow rate(kg/hr)	Molar flow rate(kmol/hr)	
Formaldehyde(HCHO)	32970.17096	1099.005699	32970.17096	1099.005699	(To E - 103)
Pentaerythritol (C5H12O4)	154862.7049	1137.441828	154862.7049	1137.441828	(To E - 104)
Sodium formate(HCOONa)	206699.5772	3039.252716	206699.5772	3039.252716	(To E - 104)
Water	154862.7049	8603.483606	116147.0287	6449.02991	(To E - 103)
			38715.67623	2149.676637	(To E - 104)
Di Pentaerythritol	23229.40574	91.35364849	23229.40574	91.35364849	(To E - 104)
Total flow rate	572624.5638	13970.5375	572624.5638	13965.76044	
Input-output	0				

Heat Exchanger (E 103)

	INPUT		OUTPUT	
Components	Mass flow rate (kg/hr)	Molar flow rate(kmol/hr)	Mass flow rate(kg/hr)	Molar flow rate(Kmol/hr)
Formaldehyde(HCHO)	32970.17096	1099.005699	32970.17096	1099.005699
Water	116147.0287	6449.02991	116147.0287	6449.02991
Total flow rate	149117.1996	7548.035609	149117.1996	7548.035609
Input-output	0			

EVAPORATOR (E 104)

	INPUT		OUTPUT	
Components	Mass flow rate (kg/hr)	Molar flow rate(kmol/hr)	Mass flow rate(kg/hr)	Molar flow rate(Kmol/hr)
Pentaerythritol (C5H12O4)	154862.7049	1137.441828	154862.7049	1137.441828
Sodium formate(HCOONa)	206699.5772	3039.252716	206699.5772	3039.252716
Water	38715.67623	2149.676637	38715.67623	2149.676637
Di Pentaerythritol	23229.40574	91.35364849	23229.40574	91.35364849
Total flow rate	423507.3641	6417.72483	423507.3641	6417.72483
Input-output	0			

CRYSTALLIZER (V 101)

Efficiency	65%			
	INPUT		OUTPUT	
Components	Mass flow rate (kg/hr)	Molar flow rate(kmol/hr)	Mass flow rate(kg/hr)	Molar flow rate(Kmol/hr)
Pentaerythritol (C5H12O4)-aq	154862.7049	1137.441828	23229.40574	170.6162742
Pentaerythritol (C5H12O4) - s	0	0	131633.2992	966.825554
Dipentaerythritol	23229.40574	91.35364849	23229.40574	91.35364849
Sodium formate(HCOONa)	206699.5772	3039.699665	206699.5772	3039.699665
Water	38715.67623	2149.676637	38715.67623	2149.676637
Total flow rate	423507.3641	6418.171779	423507.3641	6418.171779
Input-output	0			

EVAPORATOR (E 105)

	INPUT		OUTPUT	

Components	Mass flow rate (kg/hr)	Molar flow rate(kmol/hr)	Mass flow rate(kg/hr)	Molar flow rate(Kmol/hr)
Pentaerythritol (C5H12O4)-aq	23229.40574	170.6162742	23229.40574	170.6162742
Pentaerythritol (C5H12O4) - s	131633.2992	966.825554	131633.2992	966.825554
Dipentaerythritol	23229.40574	91.35364849	23229.40574	91.35364849
Sodium formate(HCOONa)	206699.5772	3039.252716	206699.5772	3039.252716
Water	38715.67623	2149.676637	38715.67623	2149.676637
Total flow rate	423507.3641	6417.72483	423507.3641	6417.72483
Input-output	0			

CENTRIFUGAL PUMP (V 102)

Efficiency	65.00%				
	INPUT		OUTPUT		
Components	Mass flow rate (kg/hr)	Molar flow rate(kmol/hr)	Mass flow rate(kg/hr)	Molar flow rate(Kmol/hr)	
Pentaerythritol (C5H12O4) - s	131633.2992	966.825554	131633.2992	966.825554	to V -103
Pentaerythritol (C5H12O4)-aq	23229.40574	170.6162742	23229.40574	170.6162742	the output
Dipentaerythritol	23229.40574	91.35364849	23229.40574	91.35364849	the output
Sodium formate(HCOONa)	206699.5772	3039.252716	206699.5772	3039.252716	the output
Water	38715.67623	2149.676637	32908.32479	1827.225141	the output
			5807.351434	322.4514955	to V -103
Total flow rate	423507.3641	6417.72483	423507.3641	6417.72483	
Input-output	0				

DRIER (V 103)

Efficiency	70%			
	INPUT		OUTPUT	
Components	Mass flow rate (kg/hr)	Molar flow rate(kmol/hr)	Mass flow rate(kg/hr)	Molar flow rate(Kmol/hr)
Pentaerythritol (C5H12O4) - s	131633.2992	966.825554	131633.2992	966.825554
Water	5807.351434	322.4514955	1742.20543	96.73544866
			4065.146004	225.7160469
Total water released from pentaerythritol using drier		4065.146004	Kg/hr	
Pentaerythritol obtained from drier		133375.5046	Kg/hr	

11. Energy Balances for the Entire Process for the Production of Pentaerythritol

Pentaerythritol		Di Pentaerythritol		Sodium Formate		Water					
A	0	A	0	A	-0.00082	A	-3196.413	D	2474.455	G	-488.7163
B	-77.6922	B	-0.4	B	0.539	B	1523.29	E	3.8555	H	-285.8305
C	0.889	C	340	C	0.0034	C	-203.606	F	-256.5478		
Cp(kJ/Kmol K) temperture change from 40 to 23)	1599.551366	cp(KJ/Kmol K)	-12.1644	Cp(kJ/Kmol K)	1.497436011	Cp(KJ/kmol)	75.3	t1	0.31315	t2	0.29615

Acetaldehyde		Teamperature required for	**Sodium Hydroxide**		Temperature required		**Formaldehyde**			**Sodium Formate**			**Water**		
A	29705	308	A	-1717.754	0.304		A	61900	373	A	0.01481	373	A	1523.29	0.3133
B	123.43	304	B	419.4837	0.308		B	28.3	393	B	83.264	393	B	-203.606	0.3773
		269898.32			0.424596096				-1454778			-1778.7246		75.6	
	in Cp (kj/kmol K)	269.89832		in Cp(kj/kmol k)	0.424596096			Value of cp in KJ/KMol K	-1454.778	Cp in kj/kmol k		-1778.7246			

Mixer (M – 100)	cpmix	7281.996493	1226.421006	-160.934902					
	moles in (Kmol /hr)	moles out(Kmol/hr)	mole fraction	Tin (K)	Tout(K)	enthalpy in(KJ/Kg K)	enthalpy out(Kj/kg k)		
Naoh	1318.807	1318.8068	0.034880535	298.15	304.15	-52683.8217	8934483.354		
Acetaldehyde	1110.107	1110.1068	0.02936072	298.15	304.15	825371.5349		Q	1732468.912
Formaldehyde	5495.028	5495.0285	0.145335564	298.15	304.15	8504667.125			
Water	29885.31	29885.307	0.790423181	298.15	304.15	-2075340.396			
Total	37809.25	37809.249							
							7202014.442		

E – 100 (Heat Exchanger):

Heat Exchanger (E 100)	35	31		
	molarflowrate (kmol/hr)	Mole fraction	Cp in KJ/kmol K	cp flow rate of steam KJ/Kmol K
NaoH	1318.806839	0.028836974	0.424596096	1.111126
HCHO	5495.028494	0.120154058	282.2392	temperature change
CH3CHO	1110.106766	0.024273547	269.89832	31
H2O	37809.24888	0.826735421	72.6	35
	45733.19098	1	100.4968105	
Total energy in(kj/hr) Q	4596039.826			
Mass flow rate of steam(in kmol)	11433.29774			

R – 100:

Reactor					
cpmix-in	7267.536815	1225.124545	-160.676332		
cpmix-out	-776.5309231	1377.116798	-182.997665		

	Cp			mole fraction in	mole fraction out
NaOH	156	1379.652	0	0.036431169	0
Formaldehyde	85	5495.028	1099.0057	0.145102057	0.033227171
Acetaldehyde	120	1110.107		0.029313547	0
Water	75.3	29885.31	29885.31	0.789153227	0.903547902
Pentaerythritol	200	0	1137.4418	0	0.034389243
Sodium formate	83	0	892.71586	0	0.026990235
U Acetaldehyde	120		61.039189	0	0.00184545
Di Pentaerythritol	300	0	91.353648	0	0.002761972
		37870.09	33075.513		
	Del Cp	342			
				Q	1040429152

Neutralizing tank (V – 100)

Neutralising tank							cpmix out
		moles in	moles out	mole fraction in	mole fraction out	Tin	Tout
Sodium formate		892.7159	2198.0114	0.019662836	0.049614079	308.15	323.15
Pentaerythritol		1137.442	1137.4418	0.025053136	0.02567463	308.15	323.15
Water		39776.36	39776.358	0.876108534	0.897842193	308.15	323.15
Formaldehyde		1099.006	1099.0057	0.024206547	0.02480704	308.15	323.15
Formic Acid		1099.006	0	0.024206547	0	308.15	298.15
Di Pentaerythritol		91.35365	91.353648	0.002012143	0.002062058	308.15	323.15
Sodium hydroxide		1305.296	0	0.028750258	0		
		45401.18	44302.171				

enthalpy in	enthalpy out	rate of reaction	enthalpy rxn		
832357.1538	-2086098.13	-666500	-1025449934		

		-920500			
		-285830			
		-469200		Q	1022531479
		-425090			
		-1572300			
		-426600			

Heat Exchanger (E 101)

Heat Exchanger (E 101)				
	Molar flow rates(kmol/hr)	Molefraction	Cp in KJ/kmol K	Cp of steam
formic acid	1099.005699	0.128233731	4.827	1.1111265
Water	7471.326665	0.871766269	75.4	temperature variation
Total	8570.332364	1	66.35016087	45
Total energy in (kj/hr) Q	568642.9311			33
Mass flow rate of steam in kmol/hr	6141258.599			
				\

Pump (P - 100)				
at 50 C				
P1	1(in atm)	101325	Pa	
P2	2(in atm)	202650	Pa	
density			Molar flow rates (kmol/hr)	mole fraction
Penterythritol	1.396	g/cm^3	1137.441828	0.06384207
Di Pentaerythritol	1.365	g/cm^3	91.35364849	0.005127476
Sodium Formate	1.92	g/cm^3	6889.985908	0.386719522
Formaldehyde	1.08	g/cm^3	1099.005699	0.061684736
Water	1	g/cc	8598.706547	0.482626196
Mixture density	1121.43	kg/m^3	17816.49363	1
P2-P1/(denisty * g)	9.210335177			
Pump head	9.210335177	m		

F-100

Fired heater											
Input	Molar flowrate	Molefraction		Bmix	Amix	x	f(x)	I	Moles		

Formaldehyde	1099.005699	0.078665957	947.5428	-112.761	0.3308	179.0821824	18.64498387	C	0.054166667	kmol/kgcoal
Pentaerythritol	1137.441828	0.081417184			0.393	259.6233204		H	0.025	kmol/kgcoal
Sodium formate	3039.252716	0.2175473						S	0.0003125	kmol/kgcoal
Water	8603.483606	0.615830537					Theoratical o2		0.378472222	kmol/kgcoal
Dipentaerythritol	91.35364849	0.006539022				Excess air	20%	N2	0.358791667	kmol/kgcoal
Total	13970.5375							Total	0.438270833	kmol/kgcoal
QProcess	72.35567952	KW					mass flowrate		0.026275147	kg/s
Qflue	2.367390719	KW						Q	2.367390719	Kw
TotalQ	74.72307024	KW								
Coal required	0.002067305	Kg/s								
Calorific value of coal	35000									

Heat Exchanger (E 101)				
	Molar flow rates(kmol/hr)	Molefraction	cpdt(KJ/Kmol)	Cp of steam
formic acid	1099.005699	0.128233731	4.827	1.1111265
Water	7471.326665	0.871766269	75.4	temperature variation
Total	8570.332364	1	66.35016087	45
Total energy in (kj/hr) Q	568642.9311			33
Mass flow rate of steam in kmol/hr	6141258.599			

E-103

E-103(Heat Exchanger)				
Input	Molar flowrate	(kmol/hr)	Mole fraction	
HCHO(g)	1099.005699		0.145601552	

H20(g)	6449.02991	0.854398448	
Tota	7548.035609		
Q	-5.172250476	Kw	

Heat Exchanger (E 104)	100C	40C		
	Molar flow rates(kmol/hr)	Mole fraction	Cpdt in KJ/kmol	Mass flow rate of heating medium(kmol/hr)
Penaterythritol (aq)	1137.441828	0.177234436	-1599.551	steam
Di Pentaerythritol	91.3536	0.014234577	-12.44	-3.06195
Sodium Formate	3039.2527	0.47357168	-5300	
Water	2149.676637	0.334959307	-75.6	
Total	6417.724765	1	-2818.925426	
Total energy in (kj/hr)	-18091087.52			
Mass flow rate of steam in kmol/hr	5908354.975			
Crystallizer (V -101)				
Efficiency	65%			
Temperature change	40 C	23 C		
	313.15	296.15		
23	296.15	303		
Components	Input molar flow rates(kmol/hr)	mole fraction	Sensible heat (KJ/kmol K)	Latent Heat(kj/kmol)
Pentaerythritol(aq)	170.6162	0.031290561	-50.05085971	5810617.048
Di Pentaerythritol	91.45435331	0.016772487	-0.062198756	
Sodium Formate	3039.699665	0.557472901	-831.805316	
Water	2150.870902	0.39446405	-29.82148221	
Pentaerythritol (s)	966.8248			
Total (only including aqueous)	5452.64112	1	-911.7398567	
Total energy(sensible+latent)	839226.8149			

Cooling water (from temperature 10 to 21 C) flow rate in kmol/hr	1013.191857			
Heat duty q in kj/hr	1191.99042			
Cp value calculation				
Cp = At^2+ B*t + c				

Heat Exchanger (E 105)	23	30		
Temperature change	296.15	303.15		
Components	Molar flow rates	Cp(KJ/Kmol K)	Molar fraction	heating medium
Penaterythritol (aq)	170.6162	159.4251999	0.026577943	Hot water
Pentaerythritol(s)	966.8248	198.01	0.150608292	50
Di Pentaerythritol	91.45435331	3.708	0.014246412	60
Sodium Formate	3039.699665	-813.897	0.47351286	
Water	2150.870902	75.3	0.335054493	
Total	6419.46592	-326.049125	1	
Total energy , Q (kj/hr)	-2093061.246			
Mass flow rate of steam(in kmol)	-2779.629809			

Heat Exchanger (E 106)				
	Molar flow rates(kmol/hr)	cpdt(KJ/Kmol)	Mole fraction	Cooling medium
Pentaerythritol(s)	966.825554	-198.2	0.967724857	Chilled Water
H20	32.24514955	-75.5	0.032275143	
Total	999.0707036	-194.23984		
Total energy (taken out)	-18435636.69	KJ/hr		

Drier (V - 103)			
	Molar flow rates(kmol/hr)	cp(KJ/Kmol K)	Molar fraction
Penaterythritol (s)	966.825554	198.2	0.749897436
Water	322.4514955	75.5	0.250102564
Total Molar flow rates	1289.27705	167.5124154	
Energy supplied to the drier(kj/s) Q	215969.9127		
divided by efficiency	254082.2503		

12. Design of Major Process Equipment

Equipment Name	**Designed by**
Rotary Dryer	Deepthisri KR
Reactor	Sivadharshini A
Evaporator	Sugavaneshwarar D
Heat Exchanger	Nived Krishna S

Dryer design :

Assumption :

Inlet air moisture content – 0.01
Peripheral speed of dryer - 0.5 m/s
Particle size (ranges between 200 micrometres to 300 micrometre) - 0.25mm
Slope of rotary dryer – 3 degree – 0.05236 radians
Gas velocity , G –20000 kg/m^2hr
Efficiency-70%
Acceptable_range:

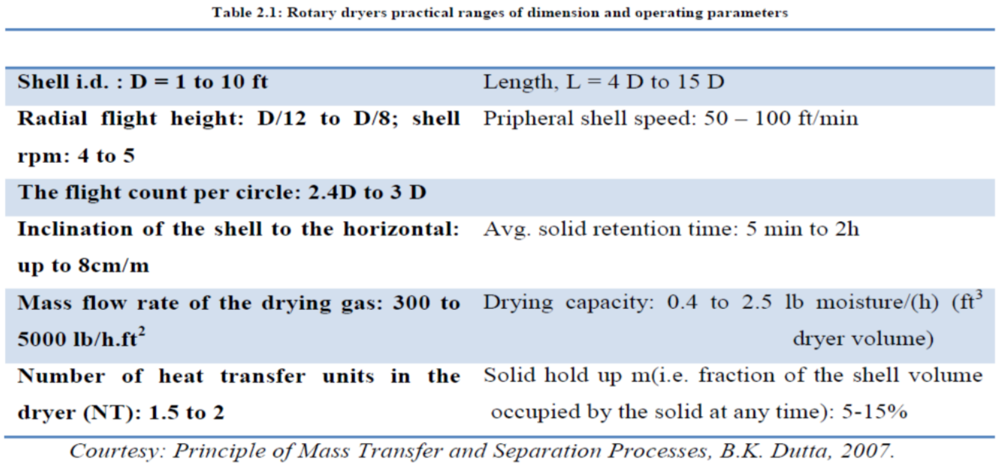


Table 2.1: Rotary dryers practical ranges of dimension and operating parameters

Shell i.d. : D = 1 to 10 ft	Length, L = 4 D to 15 D
Radial flight height: D/12 to D/8; shell rpm: 4 to 5	Pripheral shell speed: 50 – 100 ft/min
The flight count per circle: 2.4D to 3 D	
Inclination of the shell to the horizontal: up to 8cm/m	Avg. solid retention time: 5 min to 2h
Mass flow rate of the drying gas: 300 to 5000 lb/h.ft^2	Drying capacity: 0.4 to 2.5 lb moisture/(h) (ft^3 dryer volume)
Number of heat transfer units in the dryer (NT): 1.5 to 2	Solid hold up m(i.e. fraction of the shell volume occupied by the solid at any time): 5-15%

Courtesy: Principle of Mass Transfer and Separation Processes, B.K. Dutta, 2007.

Wet basis moisture content	
Initial (Xa)	0.042253521 (mass fraction formula)
Final (Xb)	0.013062409
Dry Basis Moisture Content	
Initial (Xa)	0.044117647 (moisture content/ drymass)
Final (Xb)	0.013235294

Material Balance:

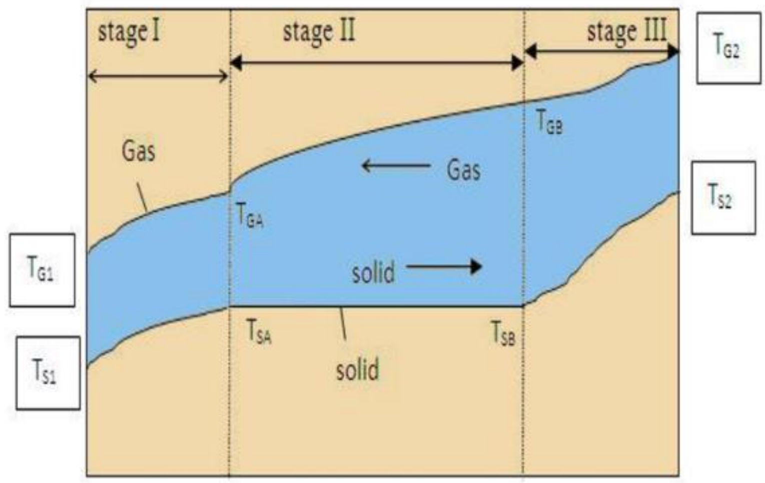
	INPUT		**OUTPUT**	
Components	Mass flow rate (kg/hr)	Molar flow rate(kmol/hr)	Mass flow rate(kg/hr)	Molar flow rate(Kmol/hr)
Pentaerythritol ($C_5H_{12}O_4$) - s	131633.2992	966.8255542	131633.2992	966.8255542
Water	5807.351434	322.4514955	1742.20543	96.73544865
			4065.146004	225.7160469

Total water released from pentaerythritol using drier		**4065.146004**	Kg/hr
Pentaerythritol obtained from drier		**133375.5046**	Kg/hr
Moisture in wet solid	**X1**	0.044117647	
Moisture in dry solid	**X2**	0.013235294	
Water evaporated	4065.146004	kg/hr	

Energy calculation:

Xa (initial moisture)	0.044117647	kg/kg dA
Xb(final moisture)	0.013235294	kg/kg dA
Tsa (feed solid)	40	C
Tsb(final solid)	100	C
Tv(vaporization)	100	C
Tvb(final vapour)	80	C
Tv	100	C
Tvb	80	C
Cp of solid(Cps)	1.5	kJ/kg K
Cp of liquid water (Cpl)	4.18	kj/kg K
Cp of water vapour(Cpv)	1.9	kj/kgk
Latent heat	2257	Kj/kg

Energy		
Heating dry solid in kj/kg	Cps(Tsb - Tsa)	90
Heating moisture to Tv	XaCpl(Tv-Tsa) in KJ/kg	11.06470588
Evaporating moisture	(Xa-Xb)(lambda) in kj/kg	69.70147057
Heating residual moisture	XbCpl(Tsb-Tv)	0
Superheating moisture in kj/kg	((Xa-Xb)Cpv(Tvb-Tv)	-1.173529412
	kj/kg	169.592647
	kj	22324039.65


stage I
stage II
stage III
T_{G2}
T_{GB}
Gas
Gas
T_{S2}
T_{GA}
T_{G1}
solid
T_{SA}
T_{SB}
solid
T_{S1}

	Inlet temperature	Outlet Temperature	
Air(Tg1,Tg2)	150	80	
Solid(Ts1,Ts2)	30	120	
Humidity of air in the inlet,Y1	0.01		
Humidity of air in the outlet ,Y2			
Y2=Y1+(me/ma)	0.031317756	safe as less than 0.05	
me- mass of liquid evaporated	1.129207223	kg/s	
ma-mass of air	52.97026722	kg/s	

Logarithimic men temperature			
DEL T1	Tha-Tsb	30	C
DEL T2	Thb-Tsa	50	C
LMTD T	39.15230378	C	
Number of Transfer Units			
NTU	ln((Tha-Twa)/(Thb-Twa))		Twa - wet bulb temperature
	0.875468737		

Ha=Y1	0.01	kg water /kg dry sir
Csa	0.24+0.45*Ha	
	0.2445	Kcal / kg C
mg =qT /(1+Ha)Csa (Tha −Thb)		
Kcal/hr conversion	5340679.342	
mg	308957.1329	kg/hr
Total moist air inlet	312046.7042	kg/hr
A = ma/G	9.534648099	m^2
d = sqrt(4ma/piG)	3.484234637	m

Ua	kG^n/D	
	874.2641994	KJ/hr m^3 K
L	Q/(pi*D^2*Ua*(del T)lm)	
	17.10047243	m
L/D (ratio)	4.907956613	
V= A*L(Volume of dryer)	163.0469869	m^3
Rotational speed (rpm) (P/piD)-N	5.481431408	rpm
Residence time by friedmann and Marshall equation	22.5760024	
		B=5*dp^-0.5
		10
Dryer Diameter calculation		
A in m^2	mg(1+Hb)/G	
15.80559251		
Assume G	20000	kg/m^2 hr

Vsolid				
F*R/(desnity*b)	33.68806644			
Hold up Voliume	20.66156945	%		
Number of flights (Nf)= 3D	10.45270391	~ 10		
Flight height	0.43552933	m		
Q in KW	3726.458299	kw		
Ca	1.005	KJ/Kg K		
Temp difference	70			
ma-mass of air	52.97026722	kg/s		
Vh=((22.41*T)/273.15)*((1/28.97)+(H/18.02))		m^3/ kg dA		
Calculate Vh for inlet and outlet specific volume				
Vh in	1.217191654	Inlet air temperature		150
Vhout	1.050026298	Outlet air temperature		80
V(avg) m3/kg da	1.133608976	Wet bulb temperature		
V	ma*V(avg)			
	60.04757038	m^3/s		

The rotary dryer is designed to reduce the moisture content of pentaerythritol from an initial wet-basis moisture content of 4.23% to the final value of 1.31%, achieving the total evaporation of 4065.15 kg/hr. The process ensures efficient drying while maintaining a peripheral speed of 0.5m/s, and a slope of 3 degree(0.0524 radians) to facilitate the material movement design assumes gas velocity of 20,000 kg/m^2 hr and operates at 70% efficiency. Balancing energy use and practical performance.

The balance confirms that 131633.3 kg/hr of dry pentaerythritol and 5807.35 kg/hr of water enter the stream. Post drying 1742.2 kg/hr of residual moisture remains in the product with 4065.15 kg/hr of water evaporated. This aligns with dry basis moisture reduction from 0.0441 kg/kg to 0.0132 kg/kg validating the process ability to meet the target specifications.

The total energy demand for drying is calculated as 16959.26 kj/kg of water removed summing contributions from:

Heating dry solids (90 kj/kg)

Heating moisture to vaporization temperature (11.06kj/kg)

Latent heat of vaporization (69.70 kj/kg)

Superheating vapour (-1.17 kj/kg indicating heat recovery)

the total energy input required is 22,324,039.65 kj/hr, emphasizing the significance of thermal efficiency in design.

The dryer's critical dimensions include a 3.48 m diameter and 17.10m length yield a 163.05 m^3 volume and a L/D ratio of 4.91, typical for rotary dryers to ensure adequate residence time. Air flow calculations indicate an inlet mass flow rate of 308957.13 kg/hr(dry basis) with the humidity increasing from 0.01 kg/kg to 0.031 kg/kg at the outlet, safely below the 0.05 kg/kg limit to prevent condensation risks. the logarithmic mean temperature difference (LMTD = 39.15) and NTU analysis further validate the heat transfer efficiency.

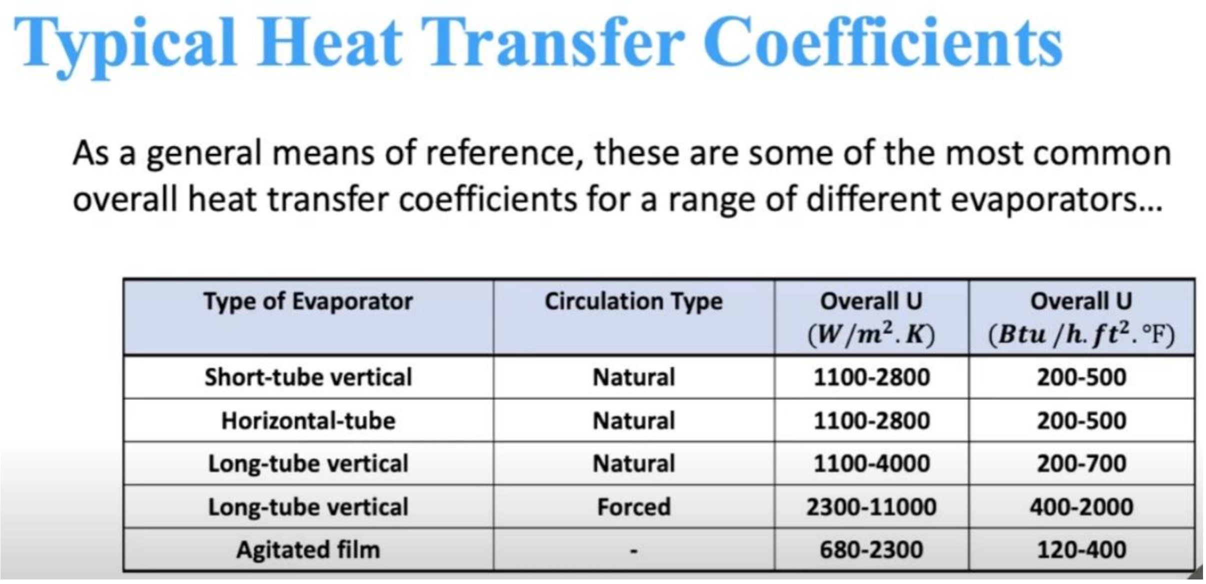
The Dryer operates at speed of 5.48 rpm ensuring optimal particle agitation and heat exposure, Residence time , calculated via Friedman and Marshal equation is 22.58 min, sufficient to achieve the desired moisture reduction for particle size (200 –300 µm).The specific volume of air is (1.13 m^3/kg dry air) and cross sectional are (9.53 m^2) ensure uniform gas solid contact.,

Outlet air humidity is 0.031 kg/kg remains within the acceptable limits,confirming operational safety. The design's reliance on a gas velocity of 20,000 kg/m^2hr and 70% efficiency reflects practical compromises between equipment size and energy consumption.

Flights are metal plates or vanes that are attached to the inside wall of the rotary dryer , They are used to lift and drop the solid particles as the drum rotates. This increases the surface area of the solid exposed to the hot air. Number of flights is calculated by 3D and the height is calculated by D/8.If there are very few and small flights, mixing is poor and the drying is very slow when there are too many flights clogging and mechanical issues might occur

Evaporator Design:

Type –Long tube vertical evaporator
Circulation - Natural



Typical Heat Transfer Coefficients

As a general means of reference, these are some of the most common overall heat transfer coefficients for a range of different evaporators…

Type of Evaporator	Circulation Type	Overall U ($W/m^2.K$)	Overall U ($Btu/h.ft^2.$°F)
Short-tube vertical	Natural	1100-2800	200-500
Horizontal-tube	Natural	1100-2800	200-500
Long-tube vertical	Natural	1100-4000	200-700
Long-tube vertical	Forced	2300-11000	400-2000
Agitated film	-	680-2300	120-400

Parameters	Values and units
Steam flow rate(S)	13970 kg/h
Overall HTC(U)	2.55 kw/m^2 k
Area , A	122 m^2
Solvent evaporated	7168.58kg/h
Steam Economy	0.51
Height,H	3.94 m
Diameter,D	1.57 m
No of tubes	6

DESIGN OF HEAT EXCHANGER E-101

INPUT	FLOWARATE Kg/hr
FORMIC ACID	1099.058
WATER	2149.678

SPECIFIC HEAT	0.75309	4.827628
THERMAL CONDUCTIVITY	0.613	0.2134
VISCOSITY	0.7973	0.8832
DENSITY	995.7	1205

Q=m*cp*dt

LMTD=dT1- dT2/Ln(dT1/dT2)

Substitute the inlet and outlet temp of the hot fluid and cold fluid

DtLMTD=34.71873

First we assume the U value find area is greater 5 so it is shell and tube

Two shell and four tube passes

Formic acid pass through tube and the water passed through the shell

Find correction factor

Plot R and P value

F=0.925

F=0.925

Then new dtLMTD VALUE = lmtd*F

= 32.11483

Find area using new LMTD value

A=1421.688

MATERIAL USED STAINLESS STELL

For tube find heat transfer coefficient U value

OUTER DIA	0.02m
INNER DIA	0.018
LENGTH	5

No of TUBE =AREA/3.14*Do*L = 452.351

VOLUMETRIC FOWRATE=Mass Flowrate/density = 376848/1205 =312.736

Per Tube = V/No of tube = 6.907

Area of Tube=3.14*Di^2/4= 2.54*10^-5

Velocity of tube = per tube / Area of tube=21069.856

Reynolds No =density*velocity*Di / viscosity = 5174457

Prandalt NO = VISCOSITY*Cp/Thermal conductivity = 19.980136

Nusselt no = 0.023/Re no^0.8*Pr no^0.4 = 3.247*10^-7

Hi= Nu*K/Di =3.845*10^-6 overall heat transfer coefficient of tube

For shell

Pitch triangular=1.25

Length=0.016

Mass flowrate=241592Kg/hr

No of tube = L^2/pitch^2 = 25.6

Area of shell flow = 3.14/4*(L^2-(No of tube*tube pitch)^2)/4 = 78.17846

Shell velocity= volumetric flow per shell/area of shell=31.03651

Hydraulic diameter=1.1*(tube pitch^2)-(0.917*Do)^2)/Do = 0.00254

Reynold no= density*dia*velocity/viscosity = 0.317214

Prandalts no= viscosity*cp*/thermal conductivity= 0.97442

Nusselts no = 0.023/Re no^0.8*Pr no^0.4 = 0.288149

Ho= Nu *k/hydraulic dia= 695.394

U =I/hi +1/h0 overall heat transfer coefficient

=261.373 J/m^2 k

REACTOR DESIGN

$$CH_3CHO < -> RCH_2CHO$$

$$RCH_2CHO < - > R_2HCCHO$$

$$R_2HCCHO < \ - > R_3CCHO$$

$$R_3CCHO < \ - > (HOCH_2)_4$$

k1	0.133492244
k2	0.093018532
k3	0.077397099
k41	0.001008292
k1o	0.006674612
k2o	0.004615423
k3o	0.003905358

Rate constant – m^3/kmol hr

$A \rightleftharpoons [k_1]\ [k2]\ I_1^x$

$I_1^x \rightleftharpoons [k3][k4]\ I2^*$

$I2^* \rightleftharpoons [k5][k6]\ I3^x$

$I_2^x \rightarrow [k7]\ P$

$r\_\{I_1^x\} = k_1[A][F] - k_2[I_1^x] = 0$

$[I_1^x] = (k_1[A][F]) / k_2$

$r_{I_2^x} = k_3[I_1^x][F] - k_4[I_2^x] = 0$

$\Rightarrow [I_2^x] = (k_3 * k_1[A][F] * [F]) / (k_2 * k_4)$

$\Rightarrow [I_2^x] = (k_3 * k_1 / k_2 * k_4) * [A][F]^2$

$r_{I_3^x} = k_5[I_2^x][F] - k_6[I_3^x] = 0$

$\Rightarrow [I_3^x] = (k_5 * k_3 * k_1 / k_2 * k_4 * k_6) * [A][F]^3$

$r_P = k_7[I_3^x][P] - k_8[P][S]$

$= k_7 * [(k_5 * k_3 * k_1 / k_2 * k_4 * k_6) * [A][F]^3]$

rate of rxn	2.596748283 kmol/m^3 hr
conc. Acetaldehyde	0.029313547 Kmol/m^3
conc. Formaldehyde	0.145102057 Kmol/m^3
volume of reactor	38.4748917 m^3

POWER CONSUMPTION:

$P = Np * \rho * N^3 * D^5$

NP –power number

ρ – density of fluid = 1200 kg/m^3

Viscosity- 0.001 Pa.s

n- speed

13. Environmental regulations and safety measures

Various regulatory bodies have classified pentaerythritol as non-hazardous including,

- Globally Harmonized System (GHS)
- U.S. Department of Transportation (DOT)
- International Maritime Dangerous Goods (IMDG)
- International Air Transport Association (IATA)
- California Proposition 6

Therefore, it does not need hazard pictograms, signal words, or specific hazard statements. However, it does appear on state right-to-know lists for Massachusetts, Pennsylvania, and New Jersey, indicating the importance of maintaining appropriate documentation and communication regarding its use in the workplace, standard chemical handling procedures are expected to be followed to ensure worker safety and prevent environmental contamination during production, storage, and use.

Pentaerythritol is not inherently flammable, it may form combustion dust when processed in solid form. Therefore, all processing equipment should be designed to minimize dust accumulation and dispersal. Adequate exhaust ventilation must be installed and maintained in areas where dust formation occurs.

For a fire involving pentaerythritol, suitable extinguishing agents include water spray, alcohol-resistant foam, dry chemical powder, or carbon dioxide are used. Fires involving this material may release hazardous combustion products, specifically carbon oxides (CO and CO_2). Firefighters must wear full protective gear including a self-contained breathing apparatus.

When handling pentaerythritol, care should be taken to prevent dust inhalation and skin or eye contact. Though the compound is considered non-irritating based on animal testing, standard protective measures should still be used.

The workplace must comply with occupational exposure limits, such as a time-weighted average (TWA) of 5 to 10 mg/m^3 for dust, as recommended by NIOSH, ACGIH, and OSHA. Safe storage is essential for maintaining product integrity and reducing hazards. Pentaerythritol should be stored in tightly closed containers in a cool, dry, and well-ventilated area. The storage environment must be free of moisture and incompatible substances such as strong oxidizers, strong acids, acid chlorides, and acid anhydrides, which can react dangerously with pentaerythritol.

In the event of accidental release or spillage, care must be taken to minimize dust dispersion. The spilled material should be swept or shoveled up carefully and collected in a sealed, labeled container for proper disposal.

Pentaerythritol has low acute toxicity and is not a carcinogen nor mutagen by regulatory agencies such as IARC, NTP, OSHA, or ACGIH. The oral LD_{50} in rats exceeds 5,110 mg/kg, indicating relatively low toxicity.

Routine hygiene practices such as washing hands after handling and before eating, drinking, or using the restroom should be followed.

From an ecological standpoint, pentaerythritol poses minimal environmental risk. It is readily biodegradable, with 83.7% degradation observed over a 28-day period under aerobic conditions.

While pentaerythritol is not considered a hazardous chemical by classification, proper precautions must still be observed during its manufacture and handling. This includes managing dust formation, ensuring adequate ventilation, wearing appropriate PPE, maintaining secure storage, and following correct procedures for accidental release and disposal. The adoption of these safety practices will help ensure the safe and efficient production of pentaerythritol.

14. Conclusion

This work followed a logical sequence in dealing with the design and analysis of an industrial process for producing pentaerythritol ($C_5H_{12}O_4$), combining market directions, principles of chemical engineering, and principles of thermodynamics to present a technically sound and economically feasible option. By means of extensive market surveys, we realized that the pentaerythritol demand is increasing rapidly in industries like alkyd resins, paints, synthetic lubricants, and flame retardants, which highlights the necessity for effective and scalable manufacturing technologies. The chosen process route includes the alkaline-catalysed condensation of formaldehyde and acetaldehyde, which is optimized for high yield (>85%) and product purity (>96%) and negligible formation of byproducts. A complete process flow diagram (PFD) was created, followed by thorough material and energy balances, and an output of 131,633 kg/hr of pentaerythritol at 96.7% purity. The project also involved the mechanical and thermal design of major equipment such as the reactor, evaporator, heat exchanger, and dryer, involving calculations by heat transfer and mass flow parameters to maximize energy recovery, heat duty, and moisture removal. Moreover, an international capacity benchmarking analysis of leading pentaerythritol manufacturers such as Perstorp, Ercros, Alder S.p.A., Chemanol, Metafrax, and Hubei Yihua was conducted to gain insights on production capacity, technology innovation, and geographical reach. This combined methodology, consisting of process design, equipment optimization, and industry benchmarking, guarantees a cost-efficient, high-performance, and scalable pentaerythritol production solution.

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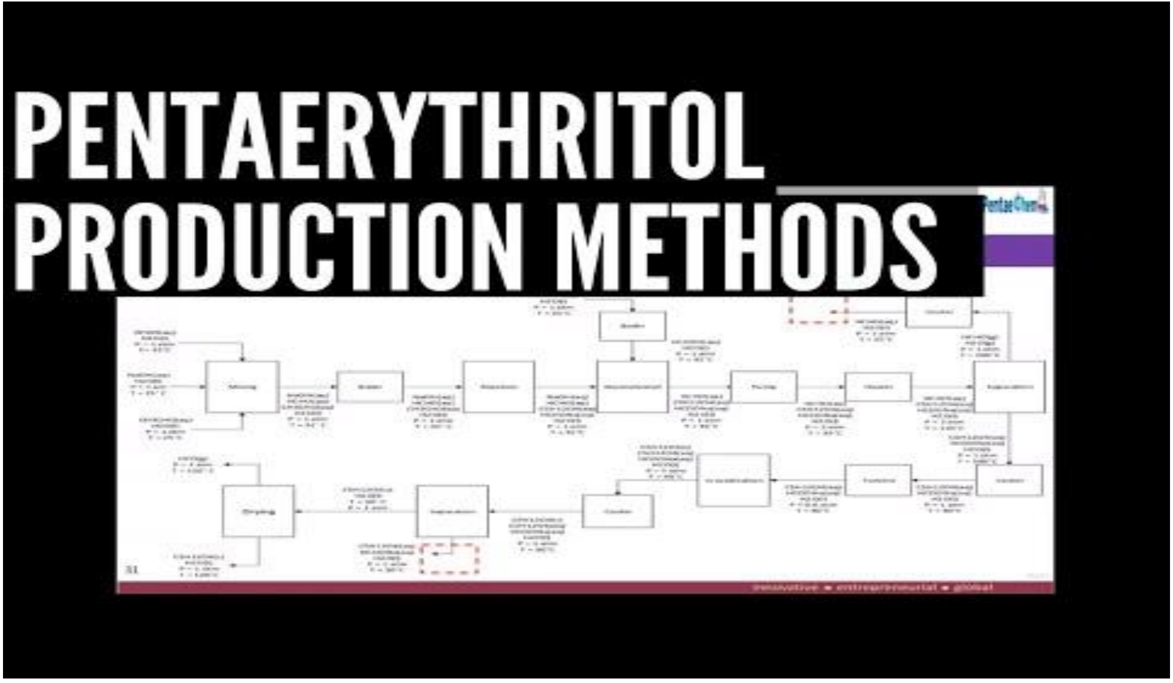

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